

Method

The model is tasked to check whether or not the nasal vowel was normal or clear speech. The architecture used was a Random Forest classifier from sklearn. The model has 1200 estimators each with a depth of 10. The labels used in training are freq f1, amp f1, width f1, freq f2, amp f2, width f2, freq f3, amp f3, width f3, amp p0, freq p0, p0prominence, a1p0, a3p0, vwl amp rms, vwl duration, appendix dur. The timepoint, clarity, and label were dropped for the training set and the only column used to check the prediction was the label column. There is a 70% 30% split on the data for the training and test sets. To check how the model is doing, calculations of accuracy, hamming loss, F1, precision, recall, macro and weighted average, and classification report.

Results

Table 1: Classification Report

type	precision	recall	f1-score	support
0 (clear)	66%	76%	71%	140
1 (normal)	76%	67%	71%	162
accuracy			71%	302
macro avg	71%	71%	71%	302
weighted avg	71%	71%	71%	302

When we look at Table 1, we see that the model levels off at around 71%. It is interesting to see that for the classification of normal speech, the precision of the model is 76%. It is hard to find why the model stabilizes at 71% but this is something to look further. I also calculated the hamming loss and accuracy. The accuracy came out to 70.86%. The hamming loss is a metric specifically for multi-class models, like this random forest. According to Scikit learn, "The Hamming loss is the fraction of labels that are incorrectly predicted." Given that, the model had a hamming loss of around 0.29 which is not desirable since we would want the loss to be as near 0

When we look at Figure 1, we see that all of the features play a role but out of the columns in the csv file, the vowel amplitude rms feature played the most into the decision tree prediction. The MDI stands for "Mean Decreased in Impurity" which, according to sklearn themselves, is computed by the mean and standard deviation of accumulation of

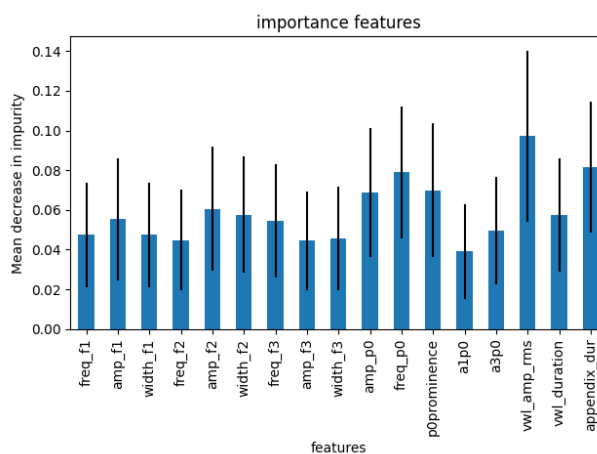


Figure 1: Feature importances using MDI

the impurity decrease within each tree. MDI calculates the total reduction of node impurity across all of the trees. There is no one feature that solely decides the random forest that does the prediction. Using the MDI, shows how each feature contributes in the decision of the prediction. The drawback with using MDI is that it can be biased towards numerical features or columns with many unique values.

Figure 2 shows the importance of the features using permutations on the full model. In this case, the features are shuffled n times, twice in Figure 2 and the model then predicts on the shuffled data. This approach is in contrast to the MDI importance, which does not take into account of biases and check each feature and their contribution. We can see the two features that are helping the model make the predictions, being the vowel amplitude rms and the appendix duration.

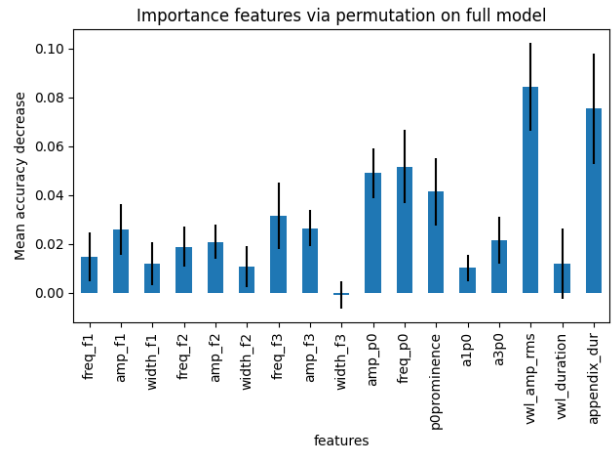


Figure 2: Feature importances using permutation